

Aggregate Pose Accuracy Is Not Invariance: Worst-Case-Over-Orbit Auditing for Robot Perception

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Abstract: Rotation robustness is normally evaluated by averaging classification accuracy over randomly sampled object poses. We argue that this average answers a different question from the one a robot poses, and we show it can conceal object-specific failures. A robot observes one realized orientation per object, determined by gravity and surrounding geometry rather than by uniform sampling. We therefore measure the empirical worst-case-over-orbit failure rate: the fraction of objects, among those a model classifies correctly at the canonical orientation, that are misclassified at one or more of the tested rotations. On a standard point-cloud benchmark, a rotation-augmented network has canonical and pose-averaged accuracies that differ by a fraction of a point, yet when several thousand orientations per object are evaluated it fails on at least one orientation for two of every five objects it recognizes upright. Objects that fail are typically misclassified at a large fraction of the tested orientations rather than at just one or two. A control network that is exactly invariant by construction, run through the same pipeline, fails on essentially none, which supports the conclusion that the measured failures come from the models rather than from the audit. For group-convolutional networks over finite rotation subgroups, increasing the group order reduces the failure rate with diminishing returns in our setting. Training on the worst of a small batch of sampled poses lowers the rate substantially at no measurable cost in canonical accuracy, but does not eliminate it. We give an audit protocol and recommend that the worst-case rate be reported beside the mean.

I. THE AVERAGE POSE IS NOT THE POSE YOU GET

Picture a picking cell. A mug tumbles down a chute and comes to rest on its side, handle pointing away from the camera. The perception module was built from rotation-equivariant features, and its paper reports that accuracy barely changes when objects are rotated, so the mug should be recognized. Suppose it is not. How surprised should we be? The reported number does not say.

It does not say because of a mismatch between evaluation and use. The evaluator draws orientations at random and averages. The robot observes one realized orientation, determined by gravity, the chute geometry and the previous action rather than by uniform sampling. If the model fails only near one stable resting orientation, the global mean over $SO(3)$ may move by less than a point while the failure repeats every time the object settles that way.

Averaging over objects compounds this. A model that fails at every orientation of one percent of objects has roughly the same aggregate error as one that fails at one percent of orientations for every object, but the two are very different to deploy: the first needs a handful of parts fixtured, the second

needs a better model. The reported statistic does not separate them.

This paper makes three contributions. First, we define the empirical worst-case-over-orbit failure rate, a per-object measure of whether any tested orientation turns a correct prediction into an incorrect one, and show what it does and does not tell you. Second, we measure it: on ModelNet40, mean accuracy over rotations can sit within a fraction of a point of canonical accuracy while a large fraction of objects fail somewhere on their orbit, and we validate this against a network that is exactly invariant by construction. Third, we give practical guidance: how to choose a pose budget, how to check that the audit itself is sound, and how much of the failure rate pose-adversarial training removes. Our evidence is 3D object recognition rather than a full manipulation stack, a limit we return to in Section VII.

II. RELATED WORK

Viewpoint robustness in non-invariant models. Alcorn *et al.* [8] found that ImageNet classifiers misclassify most of the pose space for objects they recognize when upright, and ViewFool [9] optimized adversarial viewpoints through a neural radiance field. Both search for failure poses in models that make no invariance claim, and treat pose as an attack surface.

Equivariant and invariant geometric networks. Group-equivariant CNNs [1], steerable CNNs [5], [6], tensor field networks [4] and Vector Neurons [3], surveyed under a common framework in [2], impose prescribed transformation laws on intermediate representations. Invariant classifiers built from such representations can produce pose-stable labels when the implemented symmetry matches the transformations of interest. Our finite-group models instantiate this principle on finite rotation subgroups of $SO(3)$. Dym *et al.* [12] show that, for commonly used groups, unweighted frame-averaging need not preserve continuity, and they construct weighted frames that do preserve continuity for actions including $SO(d)$ on point clouds. Their result concerns canonicalization and frame constructions rather than the empirical accuracy cost measured here.

Certification along transformation orbits. Invariance-aware randomized smoothing can augment a certified region by taking the union of that region's transformation-group orbits [10]. This construction assumes the stated model invariance holds over the transformations used; our audit instead measures how closely an implemented model satisfies that assumption.

Our focus is the evaluation statistic itself: how prevalent per-instance orbit failures are in models whose aggregate pose accuracy already looks invariant, anchored by an exactly invariant control.

III. THE WORST-CASE-OVER-ORBIT AUDIT

Let f be a classifier intended to be invariant under a compact group $\mathcal{G}$ acting on $\mathcal{X}$, let e denote the canonical pose (the orientation in which the dataset supplies the object), and let $g_1, \dots, g_M \in \mathcal{G}$ be the tested rotations. Define the eligible set and the failure rate

$$S = \{i : f(x_i) = y_i\}, \quad (1)$$

$$\text{WCO}_M = \frac{1}{|S|} \sum_{i \in S} \mathbf{1}[\exists j \leq M : f(g_j \cdot x_i) \neq y_i]. \quad (2)$$

It is an average of per-object failure indicators, where the usual statistic is an average over poses.

Two rates, and which one we report. WCO_M is empirical: it counts only failures the M tested rotations happen to reveal. A failure lying between tested poses is missed, so WCO_M underestimates the full-orbit rate WCO_∞ . It is non-decreasing in M only when the pose sets are nested: the curves in Fig. 1 are obtained by drawing subsets of a single grid and so are non-decreasing in expectation, whereas the 96- and 3072-pose grids of Table I are generated independently and are not nested, so the increase between them is observed rather than guaranteed. Every measured value we report is a WCO_M with M stated; WCO_∞ appears only as an estimate, flagged as such.

Why condition on canonical correctness. Restricting to S isolates pose-induced failures: an object counts as fragile only if the model first recognizes it and then loses that recognition under rotation. This conditioning changes the evaluated population between models, so canonical accuracy must always be reported beside WCO_M .

The mean does not determine it. Suppose a fraction τ of objects each fail on a set of orientations of measure ε/τ and the rest never fail. Mean accuracy drops by exactly ε , which can be made arbitrarily small, while WCO is τ , which can be made arbitrarily large. Without a lower bound on the size of a non-zero failure region, the mean does not determine how many objects have any failure at all.

Choosing M . If the tested poses were independent Haar samples and an object's failure region had Haar measure μ , at least one would land in it with probability $1 - (1 - \mu)^M$, so 95% detection would need $M \geq \log(0.05)/\log(1 - \mu)$: 58 poses at $\mu = 0.05$, 298 at $\mu = 0.01$ and 1496 at $\mu = 0.002$. Our sweeps use a deterministic quasi-uniform grid rather than i.i.d. samples (Section IV), so these figures are a guideline for budgeting, not an exact coverage guarantee. They still make the point that a failure region occupying a fraction of a percent of $\text{SO}(3)$ needs on the order of a thousand poses to be found reliably, and that grids of a few dozen will not find it.

Estimating the limit. If object i failed on a set of Haar measure μ_i and the M poses were independent, then

$\mathbb{E}[\text{WCO}_M] = \mathbb{E}_i[1 - (1 - \mu_i)^M]$. Modeling the μ_i as $(1 - \pi)\delta_0 + \pi \text{Beta}(a, b)$ gives $\mathbb{E}[\text{WCO}_M] = \pi(1 - B(a, b+M)/B(a, b))$, whose limit as $M \rightarrow \infty$ is π . We fit (π, a, b) by least squares subject to $\pi \geq \text{WCO}_M$ at the largest M , and bootstrap over objects for an interval. Our poses form a grid rather than independent draws, so this is an approximation; and π is identified only once the curve flattens, so we report it only then.

What WCO deliberately ignores. Once an object's failure region is non-zero, WCO treats it the same whether it covers a sliver or half the orbit. It is therefore a conservative screen and should be read together with the distribution of per-object failure fractions (Fig. 2), not in place of it.

When the mean is the right number. If a cell presented an unending stream of distinct objects, each at an independent uniformly random orientation, mean pose accuracy would be exactly the expected per-pick success rate and WCO would be beside the point. Three properties of real cells break that argument. Inventory recurs, so a failing object is not averaged away over a long horizon. Resting poses are not uniform: an object released on a table occupies one of a few stable configurations, and whether those intersect a failure region has little to do with its Haar measure. And failures are correlated across retries, since nudging and re-observing moves the object a short distance in $\text{SO}(3)$.

IV. EXPERIMENTAL SETUP

We use ModelNet40 [11]: 9843 training and 2468 test point clouds, three seeds per model, reported as mean $\pm$ standard deviation. Each cloud is subsampled once to 1024 points with a fixed index set, centered, and scaled so its farthest point lies on the unit sphere; rotations are then applied directly to the point coordinates, with no re-sampling afterwards.

Test rotations form a deterministic grid, identical for every model and seed: $M/4$ Fibonacci-sphere directions crossed with four in-plane twists, giving $R(u, \varphi) = R_z(\varphi)B(u)$ with $B(u)$ the minimal rotation carrying $\hat{z}$ to u . The 96- and 3072-pose grids are built the same way but are *not* nested, since the directions differ with M ; the canonical pose belongs to neither and is evaluated separately. Worst-of- M curves draw M poses without replacement from the largest grid.

Three regimes of invariance are compared. PN-SO3 is a PointNet [7] trained with uniformly sampled $\text{SO}(3)$ rotations: learned, approximate invariance. The G -CNN rows are group-convolutional networks over finite rotation subgroups: they lift, convolve over the group, and pool over the group, so invariance to that subgroup is exact by construction, and nothing constrains the model between group elements. VN-DGCNN is a Vector Neurons network [3], exactly invariant to the continuous group by construction, and serves as the control.

We report canonical accuracy, mean accuracy over the grid, WCO_M , and a posterior-drift statistic δ : per object the largest ℓ_1 posterior change over eight random rotations, $\max_g \|p(g \cdot x) - p(x)\|_1$, reported as the median over objects (maximum 2).

model	canon. acc.	mean acc.	WCO ₉₆	WCO ₃₀₇₂	δ
<i>exact invariance to the continuous group</i>					
VN-DGCNN	0.786	0.786	0.0002	–	$2 \cdot 10^{-6}$
<i>exact invariance to a finite subgroup</i>					
G-CNN I_{60}	0.843	0.844	0.074	–	0.244
G-CNN O_{24}	0.815	0.832	0.086	–	0.349
G-CNN T_{12}	0.789	0.808	0.117	–	0.382
G-CNN C_4	0.759	0.764	0.188	–	0.403
<i>learned, approximate invariance</i>					
PN-SO3	0.606	0.604	0.295	0.402	0.286
+ pose-adv.	0.604	0.601	0.154	0.215	0.149

Table I

comparison of rotation robustness on ModelNet40 across three random seeds. Canonical accuracy and mean accuracy over the 96-pose grid are similar, but the orbit audit reveals failures that average accuracy can hide.

WCO_M is the fraction of objects classified correctly in the canonical pose that are misclassified at least once among M tested poses. δ is the median, over objects, of the largest ℓ_1 change in predicted class-probability vectors across eight random rotations; its maximum possible value is 2. A dash means that the 3072-pose sweep was not run.

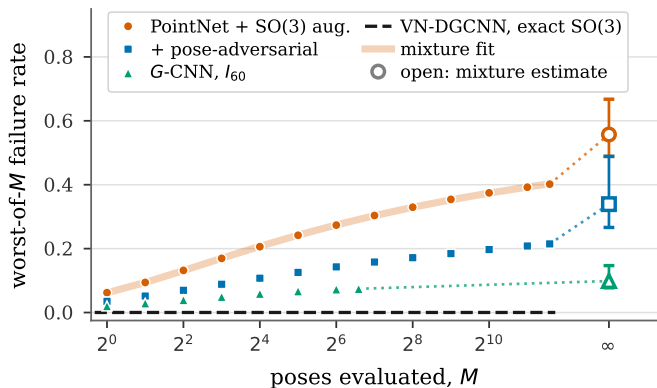


Fig. 1. Failure rate against the number of tested poses (three seeds averaged). Filled markers are measured WCO_M ; the open marker at the ∞ tick is the mixture-model estimate of WCO_∞ with its bootstrap interval, shown only for models whose curve has begun to flatten. The exactly invariant control is flat by construction.

Pose-adversarial training samples eight rotations per training example, selects the one giving the lowest probability on the true class, equivalently the largest cross-entropy, and backpropagates through that pose alone.

V. RESULTS

The mean and the count disagree. Table I gives the main result. For PN-SO3, canonical accuracy is 0.606 and mean accuracy over the grid is 0.604, an absolute difference of 0.0015. On these two numbers alone the model would be described as rotation invariant. Over the same test set, $WCO_{96} = 0.295 \pm 0.007$ and $WCO_{3072} = 0.402 \pm 0.005$. The two kinds of statistic answer different questions and we report both; we do not treat their ratio as meaningful, since one is an accuracy difference and the other a fraction of objects. Nothing is perturbed in producing the failures: every input is a valid rotation of the same clean point cloud,

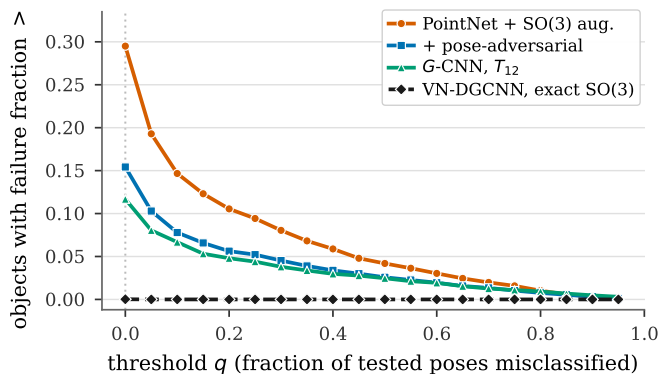


Fig. 2. Complementary cumulative distribution of the per-object failure fraction, over canonically correct objects, on the 96-pose grid. The value at $q = 0$ is WCO_{96} . Slow decay means that objects which fail do so at a large fraction of tested orientations; it says nothing about whether those orientations are contiguous in $SO(3)$.

so conventional denoising would not address the underlying pose sensitivity.

The control behaves as required. The natural worry is that we are measuring our own search. Run through the identical pipeline, VN-DGCNN returns $WCO_{96} = 0.0002 \pm 0.0002$ at a canonical accuracy higher than PN-SO3’s. This is a necessary check: had the sweep, the rotation code or prediction noise manufactured failures, they would have appeared here too. It does not by itself exclude failure modes specific to other architectures.

The measured rate depends strongly on the budget. The same model gives 0.295 at 96 poses and 0.402 at 3072, and Fig. 1 shows the curve still climbing. Its shape has flattened enough for the mixture model of Section III to be identified; the last doubling of M adds under three percent, giving $WCO_\infty \approx 0.557$ with a bootstrap interval of $[0.49, 0.67]$. We report that as an estimate, not a measurement.

Failures involve many tested poses. Among objects for which the 96-pose sweep detects at least one failure, on average about 21% of tested orientations are misclassified; conditioning on detection biases this upward. Fig. 2 shows the distribution, which has a long upper tail: a few objects per seed are recognized upright while being misclassified at more than nine tested orientations in ten. These are statements about *how much* of the tested set fails, not about its geometry. We have not measured whether the failing orientations form connected neighborhoods in $SO(3)$, so we cannot say whether a small re-orientation would recover the object; a local sweep around a detected failure would settle it.

Which objects fail. Per class, we compute for PN-SO3 the mean over instances of the fraction of the 96 tested orientations classified correctly. The lowest are *flower pot* (0.00), *cup* (0.05), *radio* (0.07) and *range hood* (0.10); the highest are *laptop*, *airplane* and *guitar*, near 1.00. Elongated, distinctive shapes are stable under rotation while open containers and boxy furniture are not. These categories resemble common manipulation targets, motivating a repeat on task-specific object sets.

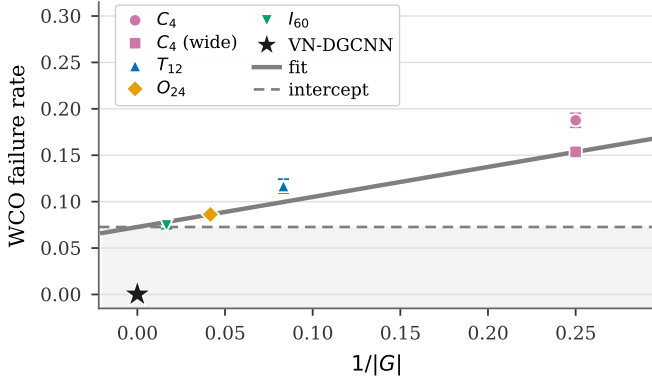


Fig. 3. Failure rate against group order (three seeds; bars are seed standard deviations). The line is fitted through the two capacity-matched configurations only. C_4 -wide and O_{24} ; T_{12} and I_{60} are shown as held-out comparisons. The dashed line marks the positive intercept of that fit. The exactly $SO(3)$ -invariant control (star, at $1/|G| = 0$) is a different architecture and is not part of the fit.

Canonical pose is not the best pose. For all four G -CNN rows mean accuracy over poses exceeds canonical accuracy, by up to 1.9 points: trained with rotation augmentation, these models spread competence over orientations, and the authored orientation is not their strongest. Evaluating only at canonical pose is not the favorable case it is assumed to be.

A. Group order: exploratory

Since equivariant pipelines usually approximate $SO(3)$ by a finite subgroup, how much a larger subgroup buys is a design question. We treat this as exploratory: it rests on four group orders within one architecture.

All four G -CNNs are exactly invariant to the group they were built for; median posterior drift over their own group lies between 3.7×10^{-7} and 2.8×10^{-6} , i.e. floating-point noise, while over the full $SO(3)$ it is 0.40, 0.38, 0.35 and 0.24. The failures live between group elements.

Group-convolution weights have shape $(|G|, C_{\text{out}}, C_{\text{in}})$, so parameters grow with $|G|$ and the ladder confounds group order with capacity. Widening C_4 to O_{24} ’s parameter count gives $WCO_{96} = 0.154$, against 0.188 narrow and 0.086 for O_{24} : the capacity-matched result recovers roughly a third of the difference, so both appear to contribute, and a single widened comparison cannot decompose them further. Fitting the two capacity-matched points gives

$$WCO_{96} = 0.073 + 0.323/|G|, \quad (3)$$

with a positive intercept, suggesting diminishing returns from increasing finite-group order. Equation 3 predicts 0.078 at $|G| = 60$ and the icosahedral model measures 0.074 ± 0.003 , within one seed standard deviation of a fit that used no $|G| = 60$ data. Whether the intercept reflects a genuine architecture-specific floor or a limitation of the group orders we could test remains open.

One methodological note: on the three smaller groups, $|G|^{-1/3}$ and $|G|^{-1/2}$ both fit with $R^2 = 0.9997$ yet miss the held-out point by 6.3 and 3.8 seed standard deviations, while $1/|G|$, with the worst in-sample fit, predicts to within

1.5. Fitting a scaling law to three points carries a model-selection risk.

VI. AN ORBIT AUDIT PROTOCOL

The following require no retraining.

- 1) Report canonical accuracy and mean accuracy over rotations, as usual.
- 2) Report WCO_M with the pose set and budget M stated, choosing M for the smallest failure region that would matter in deployment.
- 3) Plot WCO_M against M to show whether the estimate has stabilized, and extrapolate only if it has.
- 4) Run an exactly invariant control through the same pipeline. A non-zero value indicates the evaluation itself introduces apparent failures, and comparisons should wait until the source is identified.
- 5) For finite-group models, test invariance separately over G and over sampled $SO(3)$. The former should sit at the floating-point floor; ours differ by six orders of magnitude, showing the $SO(3)$ number is not an implementation artifact.
- 6) Report the distribution of per-object failure fractions alongside the binary count.

A mitigation, which does require retraining: pose-adversarial training reduces WCO_{3072} from 0.402 to 0.215 at canonical accuracy 0.604 versus 0.606, with no change outside seed noise, and lowers median posterior drift from 0.286 to 0.149. What remains is far above the control. Separately, if orbit-enlarged certificates [10] are used, let $m(x)$ be the gap between the largest and second-largest class posterior at x . If $\sup_{g \in G} \|p(g \cdot x) - p(x)\|_1 < m(x)$ then the predicted label is constant on the orbit. Our δ samples that supremum and therefore lower-bounds it, so the test can flag inputs at which the enlargement may be unsound, namely those with $\delta \geq m$, but cannot certify the remainder.

VII. LIMITATIONS AND DISCUSSION

Our evidence is object classification, not manipulation: it exercises the $SO(3)$ -invariant perception front end and nothing downstream, and we have not shown that a per-object failure rate of this size produces a comparable grasp failure rate. Test rotations form a deterministic grid rather than i.i.d. Haar samples, so the budget calculation of Section III is a guideline and the mixture estimator’s independence assumption is only approximately met. Our measurements bound how much of a tested orbit fails but say nothing about the geometry of the failing set. The control network reaches 0.786 canonical accuracy, below the published Vector Neurons ModelNet40 result [3]; we rely only on its pose response being flat, which follows from the architectural construction. Equation 3 rests on four group orders in one architecture and is exploratory.

We would like to discuss, in order of importance for robotics: whether perception-level orbit failures propagate to closed-loop manipulation failure, which is the experiment that would show whether this metric deserves attention;

how to restrict the orbit to a physically reachable and stable pose distribution, turning WCO from a stress test into a deployment estimate; whether failing orientations are contiguous, which we do not establish; and what the right quantity is for graded outputs such as grasp quality.

REFERENCES

- [1] T. S. Cohen and M. Welling, “Group equivariant convolutional networks,” in *Int. Conf. Machine Learning (ICML)*, 2016.
- [2] M. M. Bronstein, J. Bruna, T. Cohen, and P. Veličković, “Geometric deep learning: Grids, groups, graphs, geodesics, and gauges,” *arXiv:2104.13478*, 2021.
- [3] C. Deng, O. Litany, Y. Duan, A. Poulenard, A. Tagliasacchi, and L. Guibas, “Vector neurons: A general framework for $SO(3)$ -equivariant networks,” in *IEEE/CVF Int. Conf. Computer Vision (ICCV)*, 2021.
- [4] N. Thomas, T. Smidt, S. Kearnes, L. Yang, L. Li, K. Kohlhoff, and P. Riley, “Tensor field networks: Rotation- and translation-equivariant neural networks for 3D point clouds,” *arXiv:1802.08219*, 2018.
- [5] M. Weiler and G. Cesa, “General $E(2)$ -equivariant steerable CNNs,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.
- [6] M. Weiler, F. A. Hamprecht, and M. Storath, “Learning steerable filters for rotation equivariant CNNs,” in *IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2018.
- [7] C. R. Qi, H. Su, K. Mo, and L. J. Guibas, “PointNet: Deep learning on point sets for 3D classification and segmentation,” in *IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2017.
- [8] M. A. Alcorn, Q. Li, Z. Gong, C. Wang, L. Mai, W.-S. Ku, and A. Nguyen, “Strike (with) a pose: Neural networks are easily fooled by strange poses of familiar objects,” in *IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2019.
- [9] Y. Dong, S. Ruan, H. Su, C. Kang, X. Wei, and J. Zhu, “View-Fool: Evaluating the robustness of visual recognition to adversarial viewpoints,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- [10] J. Schuchardt and S. Günnemann, “Invariance-aware randomized smoothing certificates,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- [11] Z. Wu, S. Song, A. Khosla, F. Yu, L. Zhang, X. Tang, and J. Xiao, “3D ShapeNets: A deep representation for volumetric shapes,” in *IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2015.
- [12] N. Dym, H. Lawrence, and J. W. Siegel, “Equivariant frames and the impossibility of continuous canonicalization,” in *Int. Conf. Machine Learning (ICML)*, 2024.